the idea of adding behavior to the chirp so there is a whole block of variables that can be changed to affect its character, which we will explain more fully in our summary.

Listing 5-1. *Noisy Cricket Source Code*

```
const int speakerPin = 5;
const int sensorPin = 2;

const int highChirp = 20;
const int lowChirp = 14;

const int chirpCycle = 70;
const int chirpPause = 8;
const int numChirps = 10;
const int midChirp = 150;
const int skittish = 5000;

void setup() {
  pinMode(speakerPin, OUTPUT);
  pinMode(sensorPin, INPUT);
}

void loop() {
  while (digitalRead(sensorPin) == LOW) {
    for (int i=1; i<=numChirps; i++) {

      for (int j=1; j<=highChirp; j++) {
        digitalWrite(speakerPin, HIGH);
        delayMicroseconds(chirpCycle * 10);
        digitalWrite(speakerPin, LOW);
        delayMicroseconds(1000-(chirpCycle * 10));
      }

      digitalWrite(speakerPin, LOW);

      if (i == numChirps/2) delay(midChirp);
      else delay(lowChirp);

    }
    if ((random(chirpPause)) == 1) delay(random(200,1000));
    else delay(midChirp);  }
  delay(skittish);
}
```

Source Code Summary

While at first glance there seems to be quite a lot going on in this sketch, if we take the time to break it down into its various parts, it's fairly easy to understand. Let's start with the block of variables and what they do.